A Holistic Review on Detection of Malicious Browser Extensions and Links using Deep Learning

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***Abstract*—The growth of the Internet has aroused people's attention toward network security. A secure network environment is fundamental for the expeditious and impeccable development of the Internet. The majority of internet-based tasks can be completed with the help of a web browser. Although many web applications add browser extensions to improve their functionality, some of these extensions are malicious and can access sensitive data without the user's knowledge. Browser extensions with malicious intent present a growing security concern and have quickly become one of the most prevalent methods used to compromise Internet security. This is largely due to their widespread usage and the extensive privileges they possess. After being installed, these malicious extensions are executed and make an attempt to compromise the victim's browser. This makes them particularly elusive and challenging to combat. It is crucial to promptly develop an effective strategy to address the threats posed by these extensions. A comprehensive review of the research on browser extension vulnerabilities is presented in this paper. The role of malicious links in web browser extensions are examined for several attacks. Detection of malicious browser extension on various aspects are represented namely Intrusion malicious web browser extensions detection using Intrusion detection, Machine learning based detection methods and Deep learning based techniques to mitigate malicious web browser extensions are examined. This study investigates the critical function of malicious detection in protecting web browsers, looking at the changing threats and risk-reduction tactics. A robust cybersecurity frameworks can be created that not only respond to known threats but also anticipate and thwart the strategies of future cyber adversaries by realizing the significance of proactive detection. Thus this survey provides a detailed comparison of various solutions for malicious browser extension.**

***Index Terms*—Cyber-attacks, Machine learning, Malicious Browser extension, and Malicious Uniform Resource Locator (URL).**

I. Introduction

Cybersecurity is the discipline of defending data, networks, and computer systems against damage, attacks, and unauthorized access. It includes a variety of tools, procedures, and controls intended to protect digital data and guarantee the privacy, availability, and integrity of vital resources [1]. The need for strong cybersecurity strategies to combat these ever-growing challenges is highlighted by the fact that cyber threats, which range from ransomware and phishing attacks to advanced persistent threats, are becoming more and more sophisticated due to the increased sophistication of malicious actors.

The different kinds of cyber incidents and their effects on different industries [2] are depicted in the Figure 1. People can engage with all the information on the Internet via web browsers, such as Internet Explorer, Opera, Firefox, Safari, and Chrome. To make them more functional, the extensions are permitted to leverage browser features, such as browser history, bookmarks, webpage resources, and favorite social features, to enhance and personalize the user's surfing experience. Users can download a variety of extensions from third-party websites or from the official storefronts of the browser providers, such as Opera Add-Ons, Safari Extensions Gallery, and Chrome Web Store [3].

Threats posed by web browser extensions to privacy are a serious cybersecurity issue because they increase attack surfaces and make different types of exploits possible [4]. These threats—which include malware distribution, phishing, resource exploitation, and unauthorized access—highlight the necessity of implementing thorough security measures. In order to mitigate these risks and protect against ever-evolving cyber threats, careful extension vetting, user education, and proactive security strategies are necessary.

Cyber security solutions which are traditional defense mechanisms are a firewall, antivirus software or an intrusion detection system in network and computer security systems. Data science is revolutionizing in various fields, and one area where

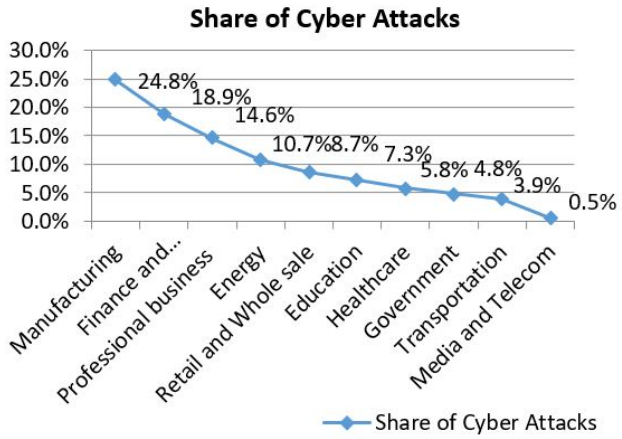

Fig. 1. Cyber Incidents by Industry

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TABLE I
RELATED WORKS

Phishing Attack [7]	The attackers may take advantage of security holes or permissions in the extension's code through Phishing emails and websites to obtain unauthorized access or steal confidential data.
SQL Injection [8]	Web pages can have malicious scripts or code injected into them by using malicious links. Clicking on such a link could cause a user with a compromised browser extension to unintentionally run the injected code, which could result in data theft, session hijacking, or other unauthorized actions on the user's part.
Drive-by downloads [9]	Drive-by download attacks, in which malicious files are automatically downloaded and installed on a user's device, can be initiated by malicious links.
Man-in-the-Browser Attacks [10]	When a user clicks on one of these links, the compromised extension might intercept and alter the user's communication with the website, which could result in theft, tampering with data, or illegal activities.
Extensions Phishing [11]	A gullible users might download and install a malicious extension, which could compromise their online accounts, sensitive data, or browser.
Social Engineering Exploitation [12]	Social engineering techniques are frequently used by malicious links to trick users.
Cross-Site Scripting (XSS) [13]	Clicking on such links may result in the execution of malicious scripts within the context of the browser extension, potentially compromising user data or session information, if the extension is not properly secured against XSS attacks.
Updates on Compromised Extensions[14]	Users who click on these links and download a compromised update run the risk of their browser extension becoming infected with malware or opening the door for more attacks on their system.

it has made a significant impact is in the realm of cyber security. With the advent of machine learning, a crucial aspect of Artificial Intelligence (AI), data scientists are now able to uncover hidden patterns and insights from vast amounts of data [5]. In the domain of cyber security, where threats are constantly evolving and becoming more sophisticated, machine learning algorithms can play a vital role in detecting and preventing potential attacks. By analyzing large datasets and identifying anomalous behavior, these algorithms can help organizations stay one step ahead of cyber criminals [6].

Deep learning methods have great potential to strengthen cybersecurity against new threats from web browser extensions. Because deep learning allows for real-time monitoring and dynamic behavioral anomaly detection, it offers a revolutionary way to combat threats posed by sophisticated web browser extensions. Using deep learning in cybersecurity offers a proactive and robust way to deal with the intricate and quickly evolving world of browser extension threats. The main aim of this paper is to show a holistic review of various machine learning and deep learning methods used to mitigate malicious browser extensions. The objective of this survey is to present a distinct goal of thoroughly examining the state of the art in cyber security research, methodologies, and advancements, with a particular emphasis on the use of deep learning techniques to recognize and mitigate the threats posed by malicious links and insecure browser extensions. The contribution of this survey examine the current obstacles and constraints on various malicious detection systems, particularly with regard to web browser extensions and links.

The organization of the literature is as follows: Section II depicts Role of Malicious Links in Web browser extension Exploitations. Section III shows Intrusion detection methods to mitigate web browser extension attacks are discussed. Section IV shows the literatures supports machine learning based technique to detect browser extension attacks. Section V depicts the state-of-art datasets, data preprocessing techniques in Deep learning techniques are examined. Section VI represents the opportunities and challenges in existing web browser extension solutions. Section VII shares the comprehensive conclusion of the study.

II. ROLE OF MALICIOUS LINKS IN WEB BROWSER EXTENSION EXPLOITATIONS

Malicious links and browser extensions are related because they can be exploited to compromise user security. Attacks can be launched through malicious links, and the dangers increase when browser extensions are used in conjunction with them. An examination of the connection between these two components is provided below as in Table I:

III. DETECTION OF MALICIOUS BROWSER EXTENSION BASED ON AN INTRUSION DETECTION METHOD

In [14] the authors examined 922,684 different malicious versions of extensions. The authors suggest a two-step method that locates the code that was added to a benign extension to turn it malicious and first distinguishes malicious extensions based on anomalous extension ratings. Some find it difficult to understand the complex euphemisms incorporated into privacy policies, while others lack the funds to pay for privacy protection services. In response to this issue, in [15] developed Secure Suite, an Internet security software package that they make available as open source, meaning it is free of cost. This malicious Uniform Resource Locator (URL) detection service is called MalURLNet. It consists of a password manager, a URL extender, data visualization tools, a browser extension for interacting with the web application, and utility tools for preserving data integrity. The purpose of intrusion detection, a crucial component of cybersecurity technology, is to track and examine network traffic from multiple sources in order to identify malicious activity. Numerous surveys of a broad range of machine learning and deep learning technique-based intrusion detection research works, including Deep Belief Networks

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(DBN), have been conducted, but they have not succeeded in providing an exhaustive analysis of all the factors pertaining to the DBN-based intrusion detection models [16]. An intrusion prevention system (IPS) is an extension of a detection system that initiates pertinent action when an intrusion is suspected in a futuristic aspect. An intrusion detection system (IDS) is a secondary intelligent system to monitor, detect, and alert about malicious activities. According to [17] this survey is the first to present a mapping technique study of risk factor analysis and to propose a hybrid framework for an effective security model for intrusion detection and/or prevention. Given the situation, intrusion detection systems (IDSs) are now an essential tool for spotting malicious activity, anomalous behaviors, and cyberattacks that jeopardize the integrity and legitimacy of networked information services. Current intrusion detection techniques that rely on feature selection are still plagued by issues such as high computational complexity or inadequate robustness. A unique ensemble feature selection-based deep neural network (EFS-DNN) is proposed by [18] to address these issues and detect attacks in networks with high volumes of traffic data.

IV. DETECTION OF MALICIOUS BROWSER EXTENSION BASED ON MACHINE LEARNING TECHNIQUES

In [19], the authors propose a novel method for classifying webpages based on keyword density and context using three supervised machine learning techniques: maximum entropy, extreme learning machine, and support vector machine. In [20] employed C4.5 as the classification algorithm after analyzing visual brand entities such as the favicon, logo, and copyright notice in phishing websites, extracting brand features and brand authorization features. In order to distinguish the site with branding rights from phishing sites, incoming links, redirection, and DNS information-based brand authorization features are further extracted to address this issue. This [21] created a feature for target identification that can recognize the target website that a phishing webpage is trying to imitate. This phishing detection system's target detection component can reduce false positives and operates faster than previously documented systems. The feature set incorporates inherent phisher limitations, making the system resistant to adaptive attackers who modify phishes to evade detection. The fundamental phishing detector component is quick and does not require internet access to centralized data. As such, it is ideal for a client-side implementation that respects privacy. The target brand identification component of this model can extract a set of keyterms that describe a webpage using a straightforward technique. If the webpage is a phish, the model use the keyterms set to identify its target. This work [22] incorporated optical character recognition for information extraction from the images and remaining other potential helpful information. This work [23] aims to develop a Chrome extension that will operate as a go-between for users and malicious websites, reducing the likelihood that users will fall victim to them. Furthermore, since harmful content is constantly evolving, it is impossible to compile all of it. In order to combat this, the authors employed machine learning to train the tool and classify newly discovered content into specific categories each time, allowing appropriate action to be performed. SVM was utilized by in [24] to find malicious URLs on Facebook. In [25], the authors used Random Forest as the classifier after extracting features from user-provided URLs and source code.

V. DETECTION OF MALICIOUS BROWSER EXTENSION BASED ON A DEEP LEARNING TECHNIQUES

A. Web Browser Attacks

A class of attacks known as "extension revelation attacks," in which extensions expose themselves by inserting their code onto websites, is examined by [26]. It offer a proof of concept implementation that prevents both types of attacks namely probing and revelation attacks as a countermeasure and suggest a browser-based mechanism that permits control over which extensions are loaded on which web pages. Small programs run in the browser context, known as browser extensions, add features and improve the online browsing experience for users. A multitude of factors, including the relative installation times of each extension, determine the order in which it operates. A multitude of factors, including the relative installation times of each extension, determine the order in which it operates.

B. Datasets

A good model must be designed with large datasets for training. The need for fresh and high-quality datasets has grown in response to the rise in phishing attacks in recent years, as model performance has to improve. Thus, in order to gain a deeper understanding of the state-of-the-art models, datasets are to be examined. Thus, benign, widely-used datasets and phishing sites are listed in Table II [27]. The following is a review of the most popular phishing detection datasets found in state-of-the-art papers.

- PhishTank [28] is the most widely used dataset for phishing detection models. Cisco Talos Intelligence Group (Talos) operated blacklist containing around 7 million phishing URL records. At present, nearly two million records are certified as phishing. Moreover, 11000 URLs of the two million are online, and the rest are offline because the phishing URL are removed whenever the attacks are performed. If it is an online URL, then it could be viewed using PhishTank.
- A data set called NSL-KDD [29] is proposed to address a few of the KDD'99 data set's intrinsic issues, which are listed in [30]. Moreover, the NSL-KDD train and test sets have a respectable amount of records.
- ISCX-URL 2016 [31]: It examine a low-tech method for identifying and classifying malicious URLs based on the nature of their attacks, and it demonstrate the efficacy and efficiency of lexical analysis in proactive detection of these URLs. It primarily examine five categories of URLs namely Benign URLs, Spam URLs, Phishing URLs, Malware URLs, and Defacement URLs.

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TABLE II
LIST OF DATASETS

S.No	Name	Last update	Type	Link
1.	PhishTank	2021	Phishing	https://phishtank.org/index.php
2.	Alexa	2021	Benign	https://www.alexa.com/topsites
3.	JoeWein	2021	Phishing	https://joewein.net/
4.	Common Crawl	2021	Benign	https://commoncrawl.org/
5.	Open Phish	2021	Phishing	https://openphish.com/
6.	Phishing Army	2021	Phishing	https://phishing.army/
7.	Kaggle	2021	Benign	https://www.kaggle.com/datasets
8.	UCI ML repository	2021	Phishing	http://archive.ics.uci.edu/
9.	Phishing -Dataset	2022	Phishing	https://github.com/GregaVrbancic/Phishing-dataset
10.	Parsed DMOZ data	2022	Benign	https://dataverse.harvard.edu/dataset.xhtml?persistentId=doi:10.7910/DVN/OMV93V
11.	Yahoo	2022	Benign	https://webscope.sandbox.yahoo.com/?guccounter=1
12.	Yandex	2022	Benign	https://yandex.com/dev/xml/
13.	Phishload	2022	Phishing	https://www.medien.ifi.lmu.de/team/max.maurer/files/phishload/

- The OpenPhish Database [32] is an archive of searchable, structured data on all the phishing websites that OpenPhish has identified. It is updated on a regular basis. The metadata in the database can also be utilized as a training or validation dataset for AI applications, as well as for the detection and analysis of cyber incidents and trend analysis.
- UCI ML repository [33]: The goal of the UCI dataset is to support network intrusion detection research. In addition to numerous cyberthreats like denial of service (DoS) attacks, malware infections, port scanning, brute force attacks, and phishing, it also contains synthetic data that mimics typical traffic. There are 100,000 records in the dataset, equally divided between malicious and legitimate activity.
- Alexa [34] is a website maintained by Amazon that provides analytic tools to analyze the performance of the websites. It performs website ranking, which is a tool that is used to rank Uniform Resource Locator (URLs) that are benign based on probability.
- Joe Wein [35] is an open-source API containing About 15000 phishing URL that are blacklist.
- Common Crawl [36] is a website which crawls over the web and suggest many datasets of legitimate website and their metadata namely page content, URL, etc.

C. Deep Learning Process

Deep Learning (DL) represents the following steps namely data collection, data preprocessing, model training and model evaluation [7].

1) Data Preprocessing: The initial step in machine learning and DL task is data preprocessing. It gets ready-made text, image, and other raw data ready for model classification. The data preprocessing used for URL-based and hybrid-based phishing detection is examined in this section. Three steps are involved in working with raw data: data cleaning, tokenization, and representation. Data cleaning first eliminates extraneous information from the input, such as special characters, stopwords, and data noise. On the other hand, a URL has special characters that stand for important information. For instance, a URL fragment is represented by the hash symbol "#." As a result, it's crucial to remove special characters carefully to avoid losing any crucial information from the URL. Therefore, the majority of URL-based models, including [37], [38], [39], and [40], do not require data cleaning.

Tokenization is an essential step in the data preprocessing process of deep learning. Tokenization is the process of dividing a text into smaller units called tokens, which are usually words or subwords. Through this procedure, raw text data can be transformed into a format that deep learning models can comprehend and handle efficiently. The input data for the URL-based can be divided into character-level segments, like [37], [39], and [41]. It can also be divided into words, like in [42]. Additionally, some papers—like [38], [40] and [43]—combine word-level and character-level analysis. Using URL symbols like "." "/" "//," and so forth, one can divide URLs into word-level segments. For model training in deep learning data preprocessing, effective data representation is crucial. Wordembedding for natural language processing, one-hot encoding for categorical variables, and normalization or standardization for numerical features are examples of common techniques. Sequence padding guarantees that the input shapes are consistent, and preprocessing for image data includes augmentation, normalization, and resizing. An increasingly stable dataset is a result of feature scaling and missing data management. The type of data and the particular requirements of the deep learning task determine which representation is best.

2) Deep Learning Modeling: Deng and Yu [44] divide deep learning techniques into two categories: generative/unsupervised models and deep discriminative models. Three methods are used in the deep discriminative models: convolutional neural networks, deep neural networks, and recurrent neural networks. Deep autoencoders, restricted Boltzmann machines, deep Boltzmann machines, and deep belief networks are the four generative/unsupervised model approaches. The broadly used deep learning algorithms for detecting malicious web browser extensions are as follows:

a) CNN:: Convolutional neural networks (CNNs) are a popular feedforward deep learning algorithm for classifying

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images. A CNN's regular architecture is made up of several layers, which are followed by the input, hidden, and output layers. Convolutional, pooling, and fully connected layers are frequently found in hidden layers [45].

b) Recurrent neural networks (RNNs):: RNNs are deep neural networks that can process text and other inputs of varying lengths because they have an internal memory function. As a result, text mining has successfully used it [46].

c) Long Short-Term Memory (LSTM) models [47]: is used in deep learning in solving problems such as the vanishing gradient problem and capturing long-term dependencies in sequential data. The fundamental elements of long short-term memory (LSTMs), such as the cell state, forget gate, input gate, and output gate. The functionality of gates is information processing and retention. Because of their ability to handle sequential data, Long Short-Term Memory (LSTM) models have found extensive applications in cybersecurity and are a good fit for a variety of security-related tasks. LSTMs are particularly useful in network intrusion detection, where they can examine network traffic patterns and identify unusual activity that could be a sign of a cyberattack. LSTMs are particularly good at detecting malware because they can analyze the temporal patterns of code execution and uncover malicious activity that may be hidden in a system's dynamic behavior. When compared to conventional methods, their capacity to maintain contextual information over long sequences improves the detection accuracy. Log analysis and cybersecurity event prediction are two applications for LSTMs. Sequential log processing enables LSTMs to spot patterns that precede security incidents and offer insights into possible threats before they materialize.

VI. OPPORTUNITIES & CHALLENGES

The research on the mitigation of web browser attacks is constantly evolving. However, there are still challenges or drawbacks in web browser extension solutions.

A. Optimal Quality datasets

The size and validity of the training dataset have a significant impact on the performance of machine learning-based solutions. The power law states that as the size of the training data increases, deep learning performance increases as well. This is contingent upon the stability of the websites or services provided by third parties [48].

B. Strategic feature extraction & selection techniques

Feature extraction and selection faces two difficulties namely: High-dimensional input spaces and Data variability. The former still have challenges associated with high-dimension data and the curse of dimensionality and there is a need for dimensionality reduction strategies. In later, it examines the difficulties caused by the distribution of data's variability and it represents the impact of noisy and irrelevant features on model performance [49].

C. Tiny URL detection

Due to the widespread usage of URL shortening services for link sharing, TinyURLs present a serious cybersecurity risk since they give attackers the chance to conceal the destination of malicious links. To effectively address the risks associated with malicious TinyURLs, a multifaceted approach combining collaborative threat intelligence, machine learning, and behavioral analysis is essential [50].

D. Real time systems and its effective responses

Rule-based models rely on third-party services from a URL string and rule parsing. As such, in a real-time prediction system that takes a single URL string as an input in every client request, they require a comparatively long response time [50]. Moreover, rule based model highly depend on heuristics and patterns which makes them least adaptable to cyber threats.

VII. CONCLUSION

This survey highlights the growing importance of DL in addressing cybersecurity challenges, specifically in the context of web browser extension attacks. It thoroughly examines how browser extensions can be manipulated to spread malicious content, underscoring the security risks they pose. The investigation reveals the sophisticated methods by which extensions can serve as vectors for cyber threats, including the dissemination of harmful links. The survey delves into various cybersecurity risks and explores detection mechanisms, with a focus on the novel integration of malicious link analysis into the exploitation of web browser extensions. This demonstrates how attacks can escalate through the combined use of malicious links and browser extensions. The survey analyzes different strategies to mitigate these risks, spotlighting the role of intrusion detection, machine learning, and deep learning techniques. It reviews extensive literature on the stages of DL processes and modeling, presenting efficient methodologies to combat cyber threats. The challenges in detecting and mitigating these risks are discussed, along with the necessity of advanced DL techniques to enhance online security. Finally, the open challenges and future directions to mitigate malicious web browser extensions are investigated. This survey is capable to elicit further discussion and research on dynamic defense to deter malicious web browser extensions.

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